

ROMAIN GERARD

French National | Software Engineering Researcher
Software Engineering (M.Eng), USTC
Suzhou, Jiangsu
[LinkedIn](#) | [Github](#) | [Website](#)
159 0560 1208 | gerarromain@gmail.com

EDUCATION

University of Science and Technology of China (USTC) **Hefei/Suzhou, China**
M.Eng. in Electronic Information - Major: Software Engineering *2025 - Present*
Coursework/Focus: Information Security, Compiler Engineering, Blockchain Technologies, Information Theory & Coding, Ubiquitous OS, HarmonyOS Development, Artificial Intelligence
Current GPA: 3.54/4.0 (First Semester)

University of Rouen Normandy **Rouen, France**
Associate + Bachelor of Science (B.Sc.) equivalent in Multimedia & Internet Technologies *2020 - 2024*
GPA: Top of Class – 16.07/20 (4.0 equivalent)

WORK EXPERIENCE

SYNK (Internship) **Remote**
Software Engineering Intern *January 2025 – August 2025*

- Developed and maintained internal Python tooling; contributed to code reviews within Agile sprint cycles across engineering teams.
- Built and tested secure Python components, participated in vulnerability assessments on internal tooling, and supported CI/CD pipeline maintenance via GitLab.

PROJECT EXPERIENCE

HexStrike-AI | LLM Security Research Framework | [Link](#) **Suzhou, China**
Security Researcher & Lead Developer *February 2026 - Present*
USTC Suzhou Institute – Information Security Lab | Engineering Project (Credited)
Supervisor: Prof. Guo Yan (郭燕)

- Extended a Python/Flask pentest framework exposing 150+ security tools via MCP, researching why LLM agents underutilize available tools and designing an improved architecture.
- Ran controlled experiments comparing Claude and DeepSeek across 4 AI clients (Claude.ai, RooCode, Sire, TRAE) on PicoCTF challenges, measuring tool-selection behavior across 3 variants: baseline preference, ranking compliance, and forced compliance.
- Built a 529-prompt benchmark corpus across 7 CTF categories × 3 difficulty levels × 2 LLMs × 3 clients for reproducible behavioral analysis.
- Deployed an OpenAI-compatible LLM proxy to intercept and log tool call chains across clients; defined 3 core metrics: ranking compliance rate, selection frequency bias, and decision divergence rate.

llm-redteam | LLM Adversarial Red-Teaming Harness | [Link](#) **Suzhou, China**
Solo Developer *June 2026*

- Built a Python red-teaming harness for OpenAI-compatible LLM endpoints across 3 attack categories (prompt injection, jailbreak, indirect injection via tool outputs); executed a 54-prompt corpus against llm-firewall and measured a 20.4% overall bypass rate, identifying direct injection as the weakest defense surface (27.8% bypass vs. 12.5% for indirect injection).

- Implemented two-pass scoring engine (regex heuristics + LLM judge for ambiguous cases) with HTML/JSON reporting and JSONL audit logging; hardened codebase across 3 security audit passes (17 → 1 finding), achieving 100% pass rate on 20 unit tests.

llm-firewall | LLM Prompt Security Proxy | [Link](#)

Suzhou, China

Solo Developer

May 2026

- Built a Python/Flask reverse proxy with a two-stage detection engine (regex rules across 8 attack categories + TF-IDF/Logistic Regression classifier) for OpenAI-compatible APIs; achieved 73.3% detection rate at 10% false-positive rate across 90 test cases, with 100% detection on jailbreak attempts.
- Regex layer handled 70% of detections; ML classifier caught the remaining 30% with 87% held-out accuracy and 100% precision on malicious inputs; weakest categories (data exfiltration at 40%, resource exhaustion at 60%) identified as priority targets for v2 rules.

ci-pipeline-guard | CLI Security Scanner | [Link](#)

Suzhou, China

Solo Developer

May 2026

- Built a CLI tool in Python that analyzes GitHub Actions and GitLab CI configuration files for supply chain attack vectors and security misconfigurations across 10 detection rules (5 critical/high).
- Outputs a weighted security score (CRITICAL -20, HIGH -10, MEDIUM -5) with Rich terminal table or JSON format; runs fully offline with no external dependencies.

OS Engineering Labs – Ubiquitous Computing Course

Suzhou, China

Graduate Coursework

Dec 2025

- Implemented a custom Linux kernel syscall in C using RCU locking for deadlock-safe process/group ID traversal; validated via kernel logs and /proc filesystem.
- Cross-compiled Linux 5.4.250 for x86 and ARM64 using GCC and QEMU; built minimal initramfs with BusyBox and custom init scripts.
- Built a buddy memory allocator in C with XOR-based partner indexing, block splitting, and cascading merge on free.
- Deployed openEuler 22.03 LTS on a two-node cluster; loaded HMDFS kernel module and validated 9 distributed filesystem behaviors including cross-node sync and write conflict detection.

SEED Labs – Network Security Practicum

Suzhou, China

Security Researcher / Graduate Coursework

Oct 2025 – Nov 2025

- Performed ARP cache poisoning (request, reply, gratuitous) in Docker environments using Scapy to enable MITM attacks on Telnet and Netcat sessions with live traffic modification.
- Exploited 32/64-bit buffer overflows via shellcode injection and GDB-assisted offset calculation, achieving root privilege escalation through Set-UID programs.
- Developed Linux kernel modules using Netfilter hooks to filter traffic by protocol, IP, and port; validated enforcement dynamically against live DNS and ICMP traffic.
- Executed ICMP redirect attacks to hijack victim routing tables, followed by TCP-layer payload interception and modification using Scapy.

SKILLS & LANGUAGES

Security: Nmap, Burp Suite, Wireshark, x86 Assembly, Scapy, GDB

Dev & Infra: Python, Flask, JavaScript, PHP, C, C#, Docker, Git, Linux/Bash, GitLab, CI/CD, PyYAML

Languages: French (Native), English (C1 - IELTS 7.5), Spanish (B1), Mandarin (Beginner - Studying)